

enclose.horse is NP-complete

Nathan Cho Agniv Sarkar

Abstract

`enclose.horse` asks the player to wall off a horse on a grid, scoring the region it encloses. We show that the decision version, stated as “is there a valid placement of score at least T ?,” is NP-complete by a polynomial-time reduction from the rectilinear Steiner tree problem. The construction uses only grass, water, cherries, golden apples, and bee swarms.

1 Introduction

`enclose.horse` is a daily grid puzzle. The board is a grid of *grass*, *water*, *cherries*, *golden apples*, *bee swarms*, and paired *portals*. A *horse* starts on a grass cell and occupies the cells reachable from it by up-down-left-right moves, with walls and water impassable. The player places up to B walls on grass cells to enclose the horse, meaning that its reachable region does not touch the board boundary, and then scores that region.

We count the grass square underneath a bonus tile. Thus a plain grass tile has value 1, a cherry tile has value $1 + 3 = 4$, a golden apple tile has value $1 + 10 = 11$, and a bee swarm tile has value $1 - 5 = -4$.

Theorem 1. *HORSE is NP-complete, and is NP-hard already on instances whose only tiles are grass, water, cherries, golden apples, and bee swarms.*

Membership in NP is easy. For hardness we reduce from the rectilinear Steiner tree problem. The construction turns each grid point into a small room and each grid edge into a narrow corridor. Terminal rooms carry a large positive apple value, nonterminal rooms have value zero, and corridors have value -3 . Thus any high-scoring pasture is forced to contain all terminals while using at most K corridors, which is exactly a rectilinear Steiner tree.

2 The decision problem

We model a board as a finite rectangular grid of typed cells. The types we use are water, plain grass, cherry, golden apple, and bee swarm. Plain grass has value 1, cherries have value 4, golden apples have value 11, and bee swarms have value -4 . Water has no value and is impassable.

A *wall placement* is a set S of plain grass cells. The horse’s starting cell may not be walled. The *pasture* of S is the set of cells reachable from the horse’s start by up-down-left-right moves avoiding S and water. The placement is *valid* if no boundary cell lies in the



Figure 1: Two ordinary `enclose.horse` instances. These examples illustrate the kind of board considered in the decision problem before we restrict attention to the structured instances used in the reduction.

pasture. The score is

$$\sigma(S) = \sum_{c \in P(S)} \text{val}(c),$$

where $P(S)$ is the pasture.

Problem 1 (HORSE). Given a board, a horse start, a budget $B \in \mathbb{Z}_{\geq 0}$, and a target $T \in \mathbb{Z}$, decide whether there is a valid placement S with $|S| \leq B$ and $\sigma(S) \geq T$.

3 Horse is in NP

Proposition 1. HORSE is in NP.

Proof. The placement S is a linear-size certificate. We check that $|S| \leq B$, that every cell of S is plain grass, and that the horse's cell is not in S . We then run a breadth-first search from the horse over the non-wall, non-water cells. This simultaneously certifies validity, by checking that no boundary cell is reached, and computes $\sigma(S)$. Everything is polynomial in the board size. $\square$

4 The rectilinear Steiner tree problem

Let $G_{\mathbb{Z}^2}$ be the infinite grid graph on $\mathbb{Z}^2$, with edges joining points at unit ℓ_1 distance.

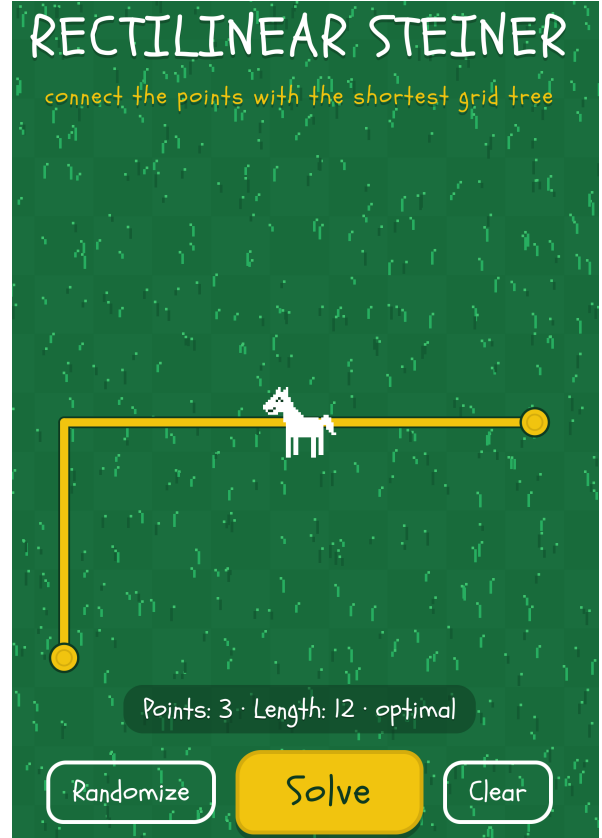
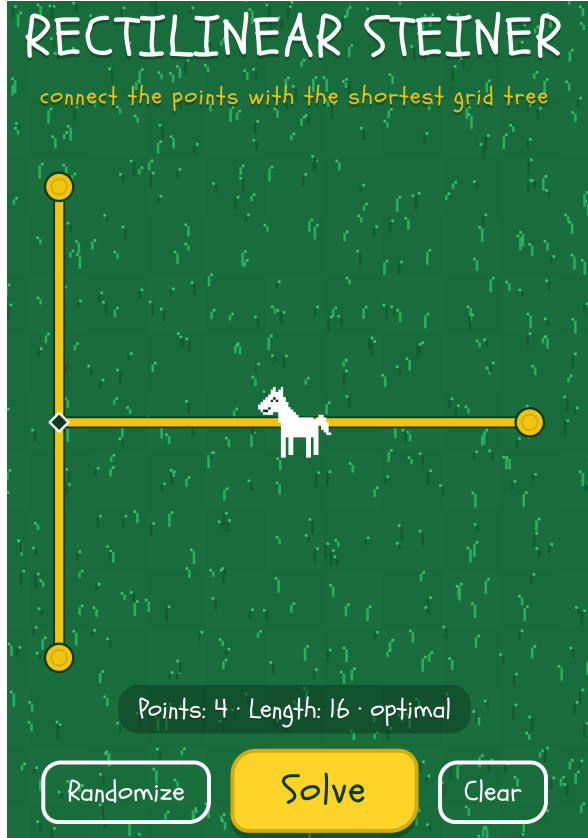


Figure 2: Two rectilinear Steiner tree witnesses before applying the reduction. The left witness is transformed into the constructed horse instance shown in Figure 5, and the right witness is transformed into the constructed horse instance shown in the same figure.

Problem 2 (RST). An instance is a tuple (R, r, K) , where $R \subseteq V(G_{\mathbb{Z}^2})$ is a finite set of terminals with polynomially bounded integer coordinates, $r \in R$ is a root terminal, and $K \in \mathbb{Z}_{\geq 0}$. It is a yes-instance if there is a connected subgraph $T \subseteq G_{\mathbb{Z}^2}$ with $R \subseteq V(T)$ and $|E(T)| \leq K$.

The polynomial coordinate bound is important, since our reduction explicitly builds a grid of comparable size. The rectilinear Steiner tree problem is NP-complete even in this strongly encoded form [GJ77]. We will use the following standard confinement fact.

Lemma 1 (Confinement to the bounding box). *If (R, r, K) is a yes-instance, then it has a witnessing subgraph contained in the axis-aligned bounding box of R .*

Proof. Clamp each coordinate to the bounding box of R . This map fixes every terminal and never increases ℓ_1 distance. Therefore it sends each unit edge either to a unit edge or to a point. Applying it to any connected witness gives a connected subgraph spanning R with no more edges than before. Taking a spanning tree if necessary gives the desired witness. $\square$

5 Why we need strong NP-completeness

The reduction below does not give a succinct description of the board. It writes down the board cell by cell. Thus the size of the board must be polynomial in the size of the input instance of RST.

This is why we use the strongly NP-complete form of rectilinear Steiner tree. In that form, the terminal coordinates are polynomially bounded in the number of terminals. Hence the bounding box of the terminals has polynomial area. After Lemma 1, it is enough to consider Steiner trees inside this bounding box, so our construction only has to build a polynomial-size portion of the integer grid.

Definition 1 (Strong NP-completeness). Let Π be a decision problem whose instances contain integers. We say that Π is *strongly NP-hard* if it remains NP-hard even after restricting to instances in which every integer parameter is bounded by a polynomial in the input length. If Π is also in NP, then Π is *strongly NP-complete*.

Equivalently, strong NP-hardness means that the hardness is not caused only by very large integers written in binary. In our setting this matters because a coordinate such as 2^n has a short binary description, but a board containing all grid points up to that coordinate would have exponential size.

Lemma 2. *In the strongly NP-complete version of RST used here, we may assume that the terminals lie in a polynomial-size bounding box.*

Proof. By the bounded-coordinate promise, if the instance has n terminals, then every terminal coordinate has absolute value at most n^c for some fixed constant c . Therefore the width and height of the terminal bounding box are each at most $2n^c + 1$. The number of integer grid points in the box is polynomial in n . $\square$

This is the only place where strong NP-completeness is used. Once the bounding box is polynomial-size, the board constructed below is also polynomial-size, since every grid point is replaced by one constant-shape room of side length $O(K)$ and every grid edge is replaced by a corridor of constant width and length three.

6 Gadgets

Let K' be the smallest odd integer strictly larger than $\max\{K, 3\}$.

All rooms fit inside a $K' \times K'$ square. Let

$$m = \frac{K' - 1}{2}.$$

The basic room shape is the plus

$$\{(i, 0) : -m \leq i \leq m\} \cup \{(0, j) : -m \leq j \leq m\}.$$

It has exactly $2K' - 1$ cells and has one port at the center of each side.

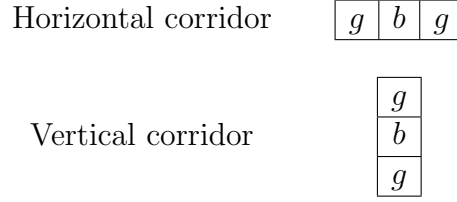


Figure 3: A corridor consists of two wallable grass gates and one bee swarm. If both gates are unwallled, the corridor contributes -2 .

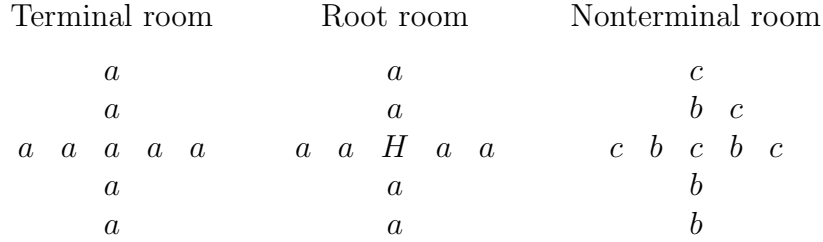


Figure 4: The three room types when $K' = 5$. Here a is a golden apple, H is the horse cell, c is a cherry, and b is a bee swarm. The nonterminal pattern is only schematic. Any placement with equally many cherries and bees works.

The corridor gadget has three cells in a line. Its two endpoint cells are plain grass and its middle cell is a bee swarm. Thus an open corridor contributes

$$1 - 4 + 1 = -2.$$

There are three room types.

A terminal room is the plus shape filled entirely with golden apples. Its value is

$$11(2K' - 1).$$

The root terminal room is the same, except that the center cell contains the horse rather than a golden apple. Since the horse stands on ordinary grass, the value of the root room is

$$11(2K' - 1) - 10.$$

A nonterminal room is the plus shape together with one extra cell adjacent to the plus. This gives $2K'$ cells. Fill exactly K' of these cells with cherries and the other K' cells with bee swarms. Its total value is

$$4K' - 4K' = 0.$$

The important features are as follows. Terminal rooms are very valuable. The root room is smaller by exactly 10, because the horse cell replaces one golden apple. Nonterminal rooms have value zero. Corridors have value -2 . The only plain grass cells in the construction, apart from the horse cell, are corridor gates.

7 Formal construction

Let (R, r, K) be an instance of RST in the strongly NP-complete bounded-coordinate form. By Lemma 1, it is enough to consider the integer grid points in the bounding box of R . Translate the instance if necessary so that this bounding box is

$$\{0, \dots, w\} \times \{0, \dots, h\}.$$

By Lemma 2, both w and h are polynomially bounded.

For every grid point in the box, place one room. If the point is the root r , place the root room. If the point is a terminal other than r , place a terminal room. Otherwise place a nonterminal room.

Place the room centers at spacing

$$K' + 3.$$

Since each room arm has length $m = (K' - 1)/2$, two neighboring rooms then have exactly three cells between their facing ports. Fill these three cells with a corridor gadget. Do this for every horizontal and vertical grid edge inside the bounding box.

All cells not used by rooms or corridors are water. Add an outer frame of water around the whole construction. The horse starts in the center of the root room.

Let M be the number of grid edges in the bounding box. Set

$$B = M$$

and

$$T = |R|11(2K' - 1) - 10 - 2K.$$

This completes the construction.

The construction is polynomial-time. The bounding box has polynomial area, each room has $O(K)$ cells, each corridor has three cells, and K' is at most $K + O(1)$.

8 Correctness

We write σ for the score on the constructed board.

First we record a simple normalization fact.

Lemma 3. *For every wall placement there is another wall placement using no more walls and having score at least as large, in which every corridor is either fully open or closed at a grass gate adjacent to the reached side.*

Proof. Consider a corridor that is only partly reached from the horse. Then the reached part consists of either one grass gate, or one grass gate together with the bee. In the first case the corridor contributes 1. In the second case it contributes -3 .

If the corridor contributes 1, leaving it as it is can only help the score, and it does not connect to a new room. If the corridor contributes -3 , wall the grass gate adjacent to the reached side instead. This uses at most one wall and removes the reached grass and bee from the pasture, increasing the score by 3.

Doing this independently for all such corridors gives the desired placement. $\square$

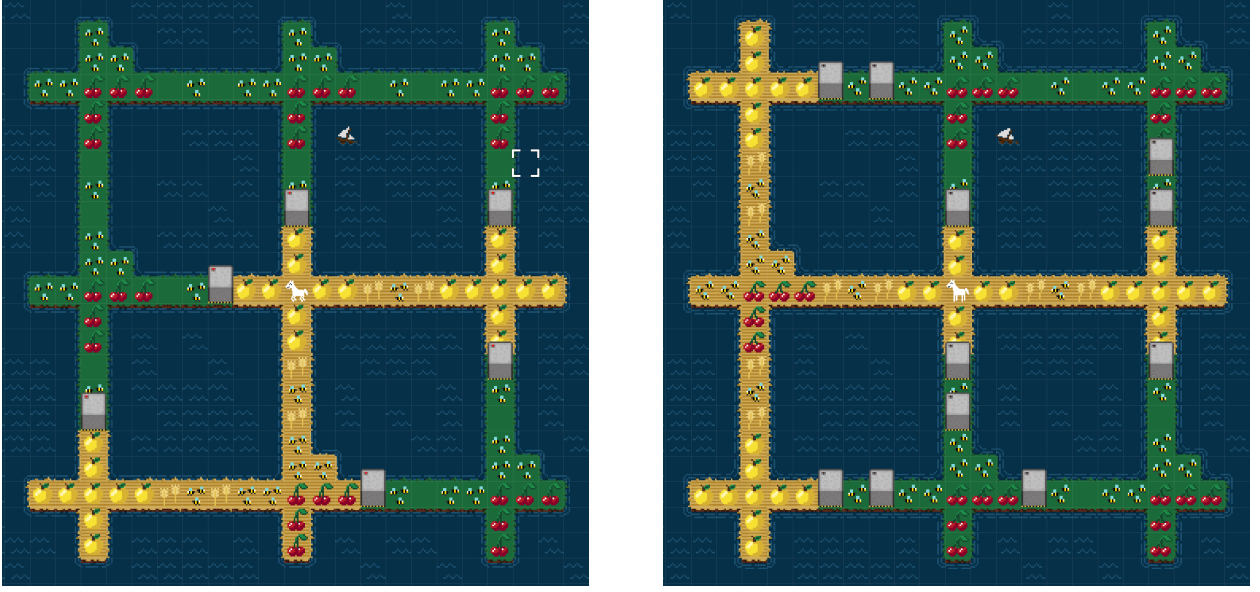


Figure 5: The horse-board instances obtained from the two Steiner witnesses in Figure 2. The left board is obtained from the left witness, and the right board is obtained from the right witness.

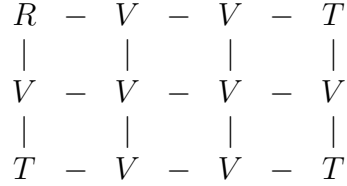


Figure 6: A schematic of the constructed board. Each R , T , or V is a room gadget. Each dash or vertical bar is a corridor gadget. The actual board is obtained by replacing this schematic grid with the rooms and corridors from Figures 3 and 4.

In the normalized placement, every reached corridor that connects two reached rooms contributes exactly -2 . Closed corridors do not help connect new rooms. Partly reached corridors may only contribute nonnegatively, and therefore they never hurt the upper bounds used below.

Lemma 4. *If (R, r, K) is a yes-instance of RST, then the constructed HORSE instance has a valid placement with score at least T .*

Proof. Let T_0 be a rectilinear Steiner tree inside the bounding box with $R \subseteq V(T_0)$ and $e \leq K$ edges. Open exactly the corridors corresponding to edges of T_0 , and wall one grass gate of every other corridor. This uses at most $M = B$ walls.

The pasture is exactly the union of the rooms and corridors corresponding to T_0 . It reaches every terminal room. The nonroot terminals contribute $(|R| - 1)11(2K' - 1)$, the root room contributes $11(2K' - 1) - 10$, and the open corridors contribute $-2e$. All nonterminal

rooms contribute zero. Hence

$$\begin{aligned}
\sigma &= (|R| - 1)11(2K' - 1) + (11(2K' - 1) - 10) - 2e \\
&= |R|11(2K' - 1) - 10 - 2e \\
&\geq |R|11(2K' - 1) - 10 - 2K \\
&= T.
\end{aligned}$$

The placement is valid because the whole construction is surrounded by water. $\square$

Lemma 5. *If the constructed HORSE instance has a valid placement with score at least T , then (R, r, K) is a yes-instance of RST.*

Proof. Let S be a valid placement with $\sigma(S) \geq T$. By Lemma 3, assume that S is normalized. The pasture determines a connected subgraph of the bounding-box grid. Its vertices are the reached rooms and its edges are the open corridors joining two reached rooms. Let e be the number of these open corridors.

We first show that every terminal room is reached. Suppose not. Since the root room is always reached, the total contribution of reached terminal rooms is at most

$$(|R| - 1)11(2K' - 1) - 10.$$

Nonterminal rooms have value zero. Open corridors have negative value. Closed corridors do not contribute. Partly reached corridors can only improve this upper bound if they contribute 1, but each such contribution is attached to a corridor that does not reach a new room, and deleting it would not decrease connectivity among reached rooms. Thus the best possible score while missing a terminal is at most

$$(|R| - 1)11(2K' - 1) - 10.$$

On the other hand, $\sigma \geq T = |R|11(2K' - 1) - 10 - 2K$. Therefore

$$(|R| - 1)11(2K' - 1) - 10 \geq |R|11(2K' - 1) - 10 - 2K,$$

so

$$11(2K' - 1) \leq 2K.$$

This is impossible because $K' > K$ and

$$11(2K' - 1) = 11(2K' - 1) > 2K.$$

Thus every terminal room is reached.

Now all terminal rooms are reached, so the score has the form

$$\sigma \leq |R|11(2K' - 1) - 10 - 2e.$$

Since $\sigma \geq |R|11(2K' - 1) - 10 - 2K$, we get

$$e \leq K.$$

The reached rooms and open corridors therefore give a connected subgraph of the bounding-box grid containing every terminal and using at most K edges. This is a feasible rectilinear Steiner tree witness for the original instance. $\square$

Proof of Theorem 1. By Proposition 1, HORSE is in NP. The construction in Section 7 is polynomial-time and uses only grass, water, cherries, golden apples, and bee swarms. By Lemmas 4 and 5, the constructed instance is a yes-instance of HORSE exactly when the original instance is a yes-instance of RST. Since RST is NP-complete in the strongly encoded bounded-coordinate form [GJ77], HORSE is NP-complete. $\square$

9 Discussion

The integer-programming approach is due to dynamight [dyn24]. However, it has a slight issue. Trinh et al. [TGD⁺26] repair it and also claim NP-completeness via size-constrained minimum s - t cut [CSS⁺16], though they do not explicitly give a gadget construction. This paper aims to fill that gap. There is an accompanying youtube video, which took a couple all nighters to film. We hope this paper and video was as fun to read and watch as it was for us to make it.

We would like to thank Prof. Li-Yang Tang and Freya Edholm for their support during CS254B!



Figure 7: Unenclosed horse...

References

- [CSS⁺16] Wenbin Chen, Nagiza F. Samatova, Matthias F. Stallmann, William Hendrix, and Weiqin Ying. On Size-Constrained Minimum s - t Cut Problems and Size-Constrained Dense Subgraph Problems. *Theoretical Computer Science*, 609:434–442, 2016.
- [dyn24] dynamight. Integer Programming Easily Encloses Horse. <https://dynamight.substack.com/p/horse>, 2024.
- [GJ77] M. R. Garey and D. S. Johnson. The Rectilinear Steiner Tree Problem is NP-Complete. *SIAM Journal on Applied Mathematics*, 32(4):826–834, 1977.
- [TGD⁺26] Kyle Trinh, Stefan Gadiant, Aston Du, Elena Tomson, and Özgür Özerdem. Solving `enclose.horse` to Optimality via Multi-Commodity Flow Integer Linear Programming Reduction. Technical report, University of California, San Diego, 2026. https://kytrinh.me/papers/eh_ilp_solver.pdf.